

Question 1 (2 points)

The union of two decidable languages is

- A. context-free but not necessarily regular
- B. regular
- C. decidable but not necessarily context-free**
- D. recognizable but not necessarily decidable

Option C is correct. Decidable languages are closed under union.

Question 2 (2 points)

Which of the following languages is not Turing-recognizable?

- A. $\{\langle R \rangle \mid R \text{ is an RE and } L(R) = \emptyset\}$
- B. $\{\langle M \rangle \mid M \text{ is a NFA and } L(M) = \emptyset\}$
- C. $\{\langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset\}$
- D. $\{\langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$**

Option D is correct. Option A, B and C are Turing recognizable. (Refer to section 16.1 in Week 7 Lecture Notes). Option D is not Turing-recognizable (Refer to section 17.5 in Week 8 Lecture Notes)

Question 3 (2 points)

Consider the language

$$L = \{\langle G_1, G_2 \rangle \mid G_1, G_2 \text{ are CFGs and } L(G_1) = L(G_2)\}.$$

Then L is

- A. regular.
- B. decidable but not regular.
- C. Turing recognizable but not decidable.
- D. not Turing recognizable.**

Option D is correct. L is not decidable (Refer to section 16.2.3 in Week 7 Lecture Notes) and so is $\bar{L}$ (because decidable languages are closed under complementation). Now, $\bar{L}$ is Turing recognizable. This is because you can construct a Turing machine M that accepts and halts if $L(G_1) \neq L(G_2)$. M just needs to test for all possible strings, whether G_1 and G_2 both can generate the string. This problem is decidable (Refer to section 16.2.1 in Week 7 Lecture Notes). If $L(G_1) \neq L(G_2)$, then M is guaranteed to find some string which is generated by one of the grammars and not by the other. But M is not guaranteed to halt if $L(G_1) = L(G_2)$. By Proposition 26 (in section 17.2.1 of Week 8 Lecture Notes) or Theorem 23 (in section 15.1 of Week 7 Lecture Notes), L is not Turing recognizable.

Question 4 (2 points)

The intersection of two Turing-recognizable languages is

- A. context-free but not necessarily regular
- B. regular
- C. decidable but not necessarily context-free
- D. recognizable but not necessarily decidable**

Option D is correct. Turing recognizable languages are closed under intersection. (Refer to Section 15.1 of Week 7 Lecture Notes)

Question 5 (2 points)

What can we say about the following two languages?

$$\begin{aligned}A_{TM} &= \{\langle M, w \rangle \mid M \text{ is a TM that accepts the string } w\} \\E_{TM} &= \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset\}\end{aligned}$$

- A. A_{TM} is Turing recognizable, E_{TM} is Turing recognizable
- B. A_{TM} is co-Turing recognizable, E_{TM} is Turing recognizable
- C. A_{TM} is Turing recognizable, E_{TM} is co-Turing recognizable**
- D. A_{TM} is co-Turing recognizable, E_{TM} is co-Turing recognizable

Option C is correct. Refer to chapter 17 (Week 8 Lecture Notes).

Question 6 (2 points)

Consider the following languages

$$\begin{aligned}L_1 &= \{\langle L \rangle \mid L \text{ is a regular language such that } |L| = 10\} \\L_2 &= \{\langle L \rangle \mid L \text{ is a regular language such that } |L| \text{ is infinity}\}\end{aligned}$$

Which of the following statement is true?

- A. L_1 and L_2 both are undecidable
- B. L_1 and L_2 both are decidable**
- C. L_1 is decidable but L_2 is undecidable
- D. L_1 is undecidable but L_2 is decidable

Option B is correct. Here, the assumption is that the description of a regular language $\langle L \rangle$ is given in the form of the DFA accepting the language. Considering the given DFA as a graph, we can compute all paths from the start state to the final states. If the number of such paths is 10, then we accept. Otherwise, we reject. Like this we can decide L_1 and hence L_1 is decidable.

Similarly, we can decide whether there is a loop across a path from the start state to some final state in the given DFA or not. If there is a loop then the cardinality of the language is infinity. Otherwise, it is not. Hence, L_2 is decidable.

Question 7 (2 points)

Which of the following statements is false?

- A. Complement of a regular language is also regular
- B. Complement of a DCFL is also a DCFL
- C. Complement of a decidable language is also decidable
- D. Complement of a Turing-recognizable language is also Turing recognizable**

Option D is correct. Refer to section 15.1 of Week 7 Lecture Notes.

Question 8 (2 points)

Consider the language

$$L = \{\langle D_1, D_2 \rangle \mid D_1, D_2 \text{ are DFAs and } L(D_1) = L(D_2)\}.$$

Then L is

- A. regular.
- B. decidable but not regular.**
- C. Turing recognizable but not decidable.
- D. not Turing recognizable.

Option B is correct. Refer to section 16.1.3 in Week 7 Lecture Notes.